

Research article

Modelling, mapping and monitoring of forest cover changes, using support vector machine, kernel logistic regression and naive bayes tree models with optical remote sensing data

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ABSTRACT

The present study is designed to monitor the spatio-temporal changes in forest cover using Remote Sensing (RS) and Geographic Information system (GIS) techniques from 1990 to 2017. Landsat data from 1990 (Thematic mapper [TM]), 2000 and 2010 (Enhanced Thematic Mapper [ETM+]), and 2013 to 2017 (Operational Land Imager/Thermal Infrared Sensor [OLI/TIRS]) were classified into the classes termed snow, water, barren land, built-up area, forest, and vegetation. The method was built using multitemporal Landsat images and the machine learning techniques Support Vector Machine (SVM), Naive Bayes Tree (NBT) and Kernel Logistic Regression (KLR). According to the results, forest area was decreased from 19,360 km² (26.0%) to 18,784 km² (25.2%) from 1990 to 2010, while forest area was increased from 18,640 km² (25.0%) to 26,765 km² (35.9%) area from 2013 to 2017 due to “One billion tree Project”. According to our findings, SVM performed better than KLR and NBT on all three accuracy metrics (recall, precision, and accuracy) and the F1 score was >0.89. The study demonstrated that concurrent reforestation in barren land areas improved methods of sustaining the forest and RS and GIS into everyday forestry organization practices in Khyber Pakhtun Khwa (KPK), Pakistan. The study results were beneficial, especially at the decision-making level for the local or provincial government of KPK and for understanding the global scenario for regional planning.

1. Introduction

Forest transformation is defined as the changes occurring in a country's forest cover from a period of contraction to a period of expansion. A forest change includes patterns from growing forest areas over time [1–3]. From a social and economic point of view,

Abbreviations: LULC, Landuse land cover; SVM, Support Vector Machine; KLR, Kernel Logistic Regression; NBT, Naive Bayes Tree.

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forests are a source of firewood and timber for construction which play an essential role in sustaining atmospheric carbon (anthropological and natural emissions) and thus contributing to mitigating the climate change [4–7]. The rapid loss of forests contributes to global climate change [8]. Logging, forestry, mining, and other human activities have changed about six million hectares (MHA) of forest worldwide [9]. According to the United Nations Framework Convention on Climate Change (UNFCCC), agriculture is the leading cause of deforestation. Commercial agriculture accounts for 32%; current agriculture accounts for 48%; forest use (infrastructure expansion such as road building and urbanization) accounts for 14%, and timber harvesting accounts for 5% of deforestation [10–12]. From an ecological point of view, forests avoid the loss of the top layers of fertile soil, which explains their importance for the entire ecosystem. The damaged canal can be visualized in deforested areas, where poor soils have significantly reduced agricultural production and hinder the potential for subsequent reforestation [6,13–16].

The forest area of Pakistan is almost 4.8 MHA in all its topographical areas [17], covering 1.98 MHA of coniferous hill forests; 1.81 MHA of foothill forests; 0.229 MHA of irrigated crops, 0.280 MHA of the river and 0.35 MHA of mangroves in the Indus River delta [18]. Pakistan's forest reservoirs are mainly located in the northern regions (6.5% in Azad Kashmir, 15.7% in Gilgit Baltistan, and 40% in Khyber Pakhtun khawa (KPK)). Forests are mainly scattered in the Karakoram, Hindukush, and Himalayan mountains and scrub in KPK. Coniferous forests are primarily located on the upper slopes of the Manshera, Mardan, Sawat, Upper Dir, Lower Dir, and Chitral Districts, where alpine pastures are rare [9].

Governments need innovative reforestation policies and programs to avoid high deforestation rates and encourage the transformation of forests. In addition to Pakistan's KPK area, some countries, such as the Republic of Korea, Vietnam, China, Japan, and the Philippines, have increased their forest cover or suffered forest change due to government action through a series of reforestation policies and programs [19–26]. Comparing these government interventions among countries is essential to provide valuable lessons to stop deforestation and improve forest coverage, especially in developing countries. The KPK's productive forest evolution through national forest policies is incredibly privileged and can be implemented in Pakistan and other developing countries to avoid deforestation and enhance their natural environments. The change recognition system assesses Land Use Land Covers (LULC) such as forest or vegetation and flood monitoring [5,27].

In previous studies [28–30], vegetation indices are considered a good indicator of the changes in the area surrounded by vegetation. To recognize the spatio-temporal evolution of vegetation indices in a semi-arid region and the relationship with the urbanization phenomenon, multi-temporal satellite image data such as MSS and ETM + have been employed previously to identify this shift using four vegetation indices (namely Transformed Soil Adjusted Vegetation Index (TSAVI), Soil Adjusted Vegetation Index (SAVI), Normalize Vegetation Difference Index (NDVI), and Transformed Vegetation Index (TVI)). The SAVI index is used to mitigate the effects of soil reflection [31]. The SAVI design is also suitable for low vegetation cover, especially in semi-arid regions. As a result, these indices could not always identify the difference between vegetation and non-vegetation regions [32]. Classification techniques can be divided into the following five groups: (a) advanced classification; (b) supervised classification; (c); unsupervised classification; (d) Object-Based Image Analysis (OBIA); and (e) sub-pixel classification. For LULC classification and mapping, numerous ML methods have been developed recently. LULC classification and forest mapping using Landsat images make extensive use of ensemble ML methods like Random Forest (RF) [33], prediction of LULC [14], forest change forecasting, and land use planning [27].

Previous researchers integrated image processing with supervised classification for mapping and classifying forest changes. There needs to be more research done locally that makes use of RS data. However, these studies [34,35], only used traditional methods like visual interpretation and traditional classifiers of optical remote sensing images. Sometimes, these methods were combined with manual digitization methods to map the forest cover and non-forest cover areas.

Extensive regional categorization and forest monitoring have been demonstrated by integrating NBT, KLR, SVM, and image differences [36]. Compared to conventional classifiers like maximum likelihood, these methods can lessen the modification and overfitting of the arrangement maps while evaluating several variables independently [37]. Accordingly, we employed the combination of KLR and SVM to extract spatial and temporal information about the KPK forests. By employing ML methods, variation and overfitting in the resulting categorized maps are reduced [35,38–41]. Several factors have aggravated deforestation and land destruction in KPK in 2010. These can be considered direct and indirect factors. The indirect factors of government officials were corruption, poor policies, lack of law enforcement, and poverty. The main factors were population growth, exploitation of fuelwood, unsustainable farming practices, illegal logging, wildland fires, and mining activities [35,15]. The reforestation took place throughout the KPK in 2014 [42]. This program was launched as part of the “One Billion Tree Project” to promote forest areas, enhance biodiversity, and monitor deforestation. A large part of the forest area was changed from 2013 to 2017 [42]. Mapping and monitoring the KPK rainforest with satellite imagery is not easy.

This research aims to monitor the forest cover change by using satellite imagery. Data were collected via Landsat using RS and GIS techniques. The information would deliver primary evidence and ground knowledge for developing a forest cover map for the “Ten Billion field project”. Consequently, this study aims to present the machine learning approach that combines SVM with KLR and NBT algorithms and uses of Landsat 5, 7 and 8 images for spatio-temporal mapping of the KPK forests, evaluate the performance of these algorithms, and put into practice a change detection strategy to track the evolution of the forest from 1990 to 2017.

2. Materials and methods

2.1. Study area

The study area selected was Khyber Pakhtun Khwa (KPK) province of Pakistan which is located between latitude 31° 40' to 36° 57' N and longitude 69° 19' to 74° 70' E in the circle of Hindukush and Himalaya Ranges, which are the world's top mountain ranges

(Fig. 1). The Hindukush range stretches along the western border, while the Himalayas stretches from North to East [43]. Many climate changes in the province were caused by significant changes in physiological characteristics and altitude. Therefore, the southern parts are dry and arid, while the northern parts are cold and humid. The northern mountains form a cold desert of high altitudes. In its central position, the Peshawar Valley has temperate weather. The average annual temperature in the south varies as per different seasons [44].

2.2. Remote sensing data and processing

In this study, we used Landsat 5 (TM), 7 (ETM+), and 8 (OLI/TIRS) RS data. We relied on Landsat images since they were freely available, cost nothing, had good spatio-temporal and spectral resolutions, and came in handy time series [18]. In addition, Landsat images with less cloud cover are given in orthorectification format [45]. Landsat 5, 7, and 8 images with 30 m spatial and 16 days temporal resolution were used [46]. Digital Elevation Model data (DEMs; 30 m $\times$ 30 m) provided by the NASA Shuttle Radar Topographic Mission (SRTM, <http://srtm.csi.cgiar.org/>). The slope was expressed in degrees, while the aspect varied from 1 to 360.

Twenty-seven years (1990, 2000, 2010, and 2013 to 2017), Landsat data were obtained from the United States Geological Survey (USGS) web engine (<http://earthexplorer.gov/>) (see Table 1). Landsat data were analyzed using SVM, NBT, and KLR algorithms. Their respective training datasets were used to compare the textural features (forests and non-forest regions) and to visually verify the obtained forest maps.

The data processing strategy was divided into two major parts. The first part of the study identified the spatio-temporal distribution of forests, which included the following steps: 1) Download Landsat data from the USGS for the years (1990, 2000, 2010, and 2013 to 2017). All the images had less than 10% cloud cover. 2) For atmospheric correction, we used the Fast Line-of-Sight Atmospheric Analysis of Spectral Hypercubes (FLAASH) module in ENVI v_5.4 [47–50]. FLAASH method consist of radiometric calibration and dark subtraction. Every Digit Number (DN) was transformed into a Top Of Atmosphere (TOA) reflectance through the process of radiometric calibration, beta nought calibration. The calibration type was reflectance, the output interleave was BSQ, the output data type was float, and the scale factor was set to 1 for this TOA calculation. Band minimum conversion of time-averaged optical attenuation (TOA)

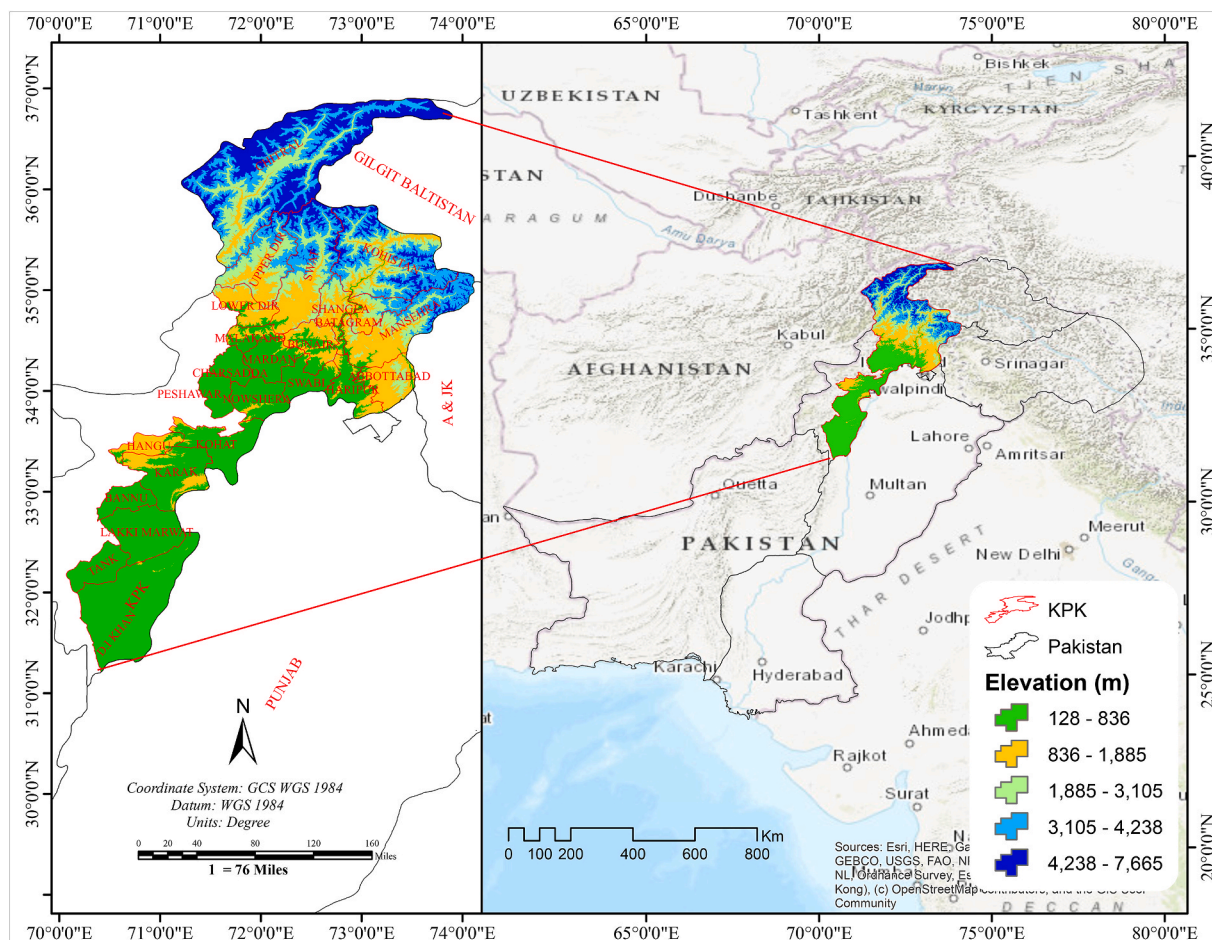


Fig. 1. Location of the study area with elevation in meters.

Table 1

Datasets were used in the study.

S. No	Acquisition Date	MOSAIC		Sensor/Product	Source	Purpose	Processing Level
		Path	Row				
01	May 1990	150 and 151	34,35,36,37 and 38	TM	USGS-EROS	LULC	TIER 1
02	May 2000	150 and 151	34,35,36,37 and 38	ETM+			
03	May 2010	150 and 151	34,35,36,37 and 38	ETM+			
04	May 2013	150 and 151	34,35,36,37 and 38	OLI/TIRS			
05	May 2014	150 and 151	34,35,36,37 and 38	OLI/TIRS			
06	May 2015	150 and 151	34,35,36,37 and 38	OLI/TIRS			
07	May 2016	150 and 151	34,35,36,37 and 38	OLI/TIRS			
08	May 2017	150 and 151	34,35,36,37 and 38	OLI/TIRS			
09	2018			SRTM	SRTM	Elevation Map	

to Surface Reflectance (SR) was used in dark object subtraction [51,52]. 3) Layer stacking and mosaicking after atmospheric correction in ERDAS Imagine 2015.4) Subset study area in Arc GIS (Arc Map) 10.8.4) Subset of shapefile in Arc GIS 10.8 using raster calculator. 5) Land cover classification mapping in Erdas imagine 2015.6) Statistical calculation of forest cover change in ENVI v.5.4.7). We used GPS for field survey and comparatively high-resolution Google Earth images for accuracy assessment in ARC GIS 10.8. Finally, we used forest cover mapping and calculation of change detection from 1990 to 2017 in Arc Map 10.8. All methodology was explained in Fig. 2.

2.3. Machine learning for image classification

2.3.1. Support vector machine

For advanced machine learning tasks, the SVM provides a suite of highly effective algorithms [53]. The goal of SVMs is to design an ideal hyperplane, also known as a decision boundary, which exploits the distance between the nearest samples (support vectors) to the plane and neatly divides classes [54–56]. The model prioritizes training instances that occur at the class distribution's periphery the support vectors while effectively discarding the rest [53]. One of the benefits of using the algorithm is that it can achieve high accuracy even with very small training datasets, which helps to reduce the overall cost of training data gathering. The SVM method of classification is based on the assumption that only training samples located on the class boundaries are necessary to make accurate distinctions [55].

The way the training samples are spread out has a big effect on the mathematical definition and design of the optimum hyperplane, i.e. whether the datasets can be separated well or only with some errors. To further regularize/compensate for misclassification/errors, the slack variable I and penalty value (C), also known as cost value, are included in cases of indiscernible samples to punish the outliers. Finally, for purely separable samples, the optimum hyperplane is determined by minimizing the norm (w) function $F(w)$, equation (1);

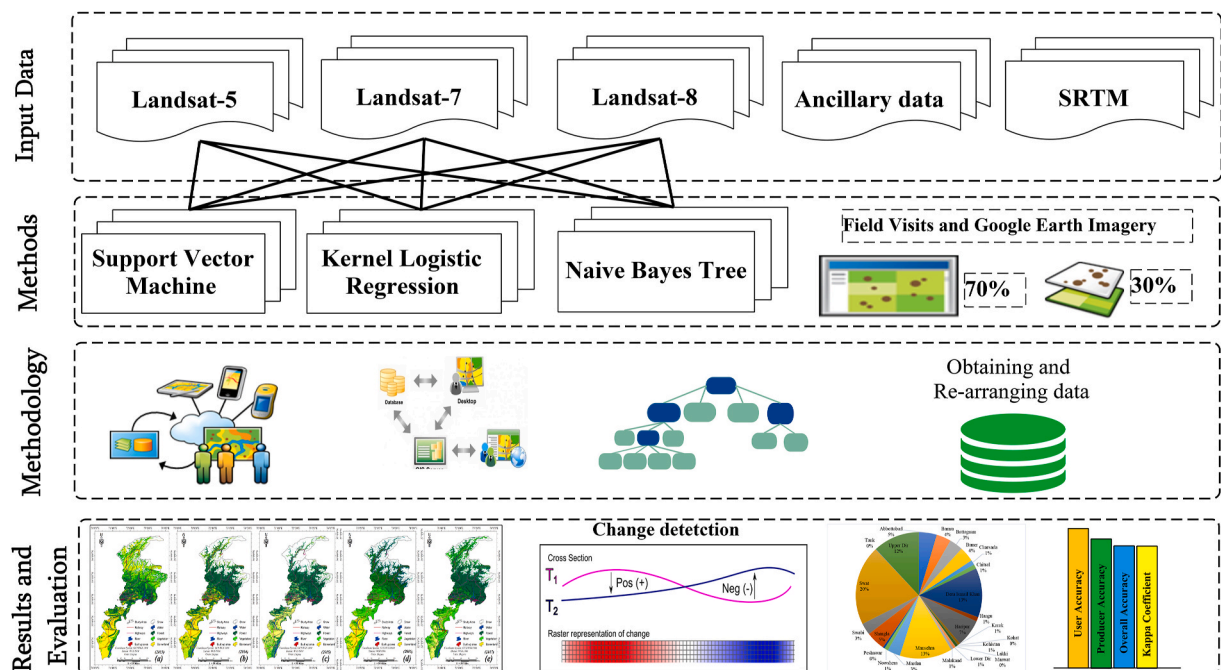


Fig. 2. Flow chart explain the methodology of the study (Khyber Pakhtun Khwa Province, Pakistan).

and for $F(w)$, equation (2), for non-differentiable data, which is the most prevalent form of data in remote sensing [55,57] Consequently, we may write them in mathematical notation as [55]:

$$F(W) = (ww)^{1/2} \quad (1)$$

$$F(W, \xi) = (ww)^{1/2} + C \left(\sum_{i=1}^k \xi_i \right)^1 \quad (2)$$

Accuracy and/or generalizability are heavily influenced by misclassified training datasets, in which the data sit on the incorrect side of the ideal hyperplane [55,57], and the variable C is used to mechanism the level of the consequence to regularize such datasets in another function (equation (2)). Overfitting and reduced ability to generalize unknown data result from a high penalty factor, which is achieved by setting C to a large number. Alternatively, if it is too small, it can lead to smoothing, which is the same as having a biased model or underfitting [55,57]. Because of this, it is essential that this parameter be set to the correct value [58]. recommends a middle-of-the-road value in this regard as a means of resolving the tension between the two alternatives [53]. They include the parameters of the kernel functions used to project/map samples that cannot be classified linearly to a higher dimensional space where they can be classed linearly, and the type of kernel functions used to do so [55].

The present study classified twenty-seven-year data collected from Landsat 5, 7, and 8 to calculate the snow, water, barren land, built-up area, forest area, and natural vegetation/shrubs from 1990 to 2017. This study analyzed LULC changes in the 1990–2010 (20 years period) and 2013–2017 (5 years period). The forest area was given due importance as our main objective was to calculate the changes in the forest area that occurred from 1990-to 2017. SVM performed a similar type of analysis, which distinguished homogeneous representative samples of the different kinds of interest on the surface cover. The classification of image data was generated using trained sites, then used to suit the pixels found by the SVM.

2.3.2. Kernel logistic regression

Discriminative ML classifier Kernel Logistic Regression (KLR) uses the Broyden-Fletcher-Goldfarb-Shanno (BFGS) optimization technique to produce probabilistic output that is then used to distinguish between the forests and non-forests classes [55]. As its name implies, this kind of logistic regression uses kernel functions to map the input feature space onto a higher-dimensional feature space through following equation (3) [26].

The kernel function is the fundamental function in which ϕ is assumed that is unknown.

$$K(x, x) = \phi(x)^T(x) \quad (3)$$

where T is the inner product in the Z space. Suppose a set of training dataset $\{x_i, y_i\}_{i=1}^{N_i-1}$

With $x_i \in \mathbb{R}^n$ as input parameters with n variables and N data samples. Here, the input parameters are dense forests, dispersed forests and non forests.

$Y_k \in \{1, 0\}$ is the related label that indicates forest and non-forests classes. The KLR was applied to built a non-linear decision boundary that may divide the two classes in the feature space based on the following equation (4):

$$p(x) = \frac{e^{y(x)}}{(1 + e^{y(x)})} = \sum_k^{N_i=1} \alpha_i K(x_i, x_j) + b \quad (4)$$

where $y(x)$ is the logistic function with values in $[0, 1]$; α_i is a vector of dual model parameters, where as b is the intercept; $K(x_i, x_j)$ is the kernel function. For our study, Radial Basis Function (RBF), which is used widely in the literature, was chosen due its flexibility and reliability [59].

2.3.3. Naïve bayes tree

Machine learning classifier Naive Bayes Tree (NBT) builds a model based on probabilities. Naive Bayes, an application of Bayes's Theorem, makes the system function. The NB bases its framework off on a Decision Tree (DT), with the NBT model being precisely laid out at each branch's terminal node. The NBT has exceptional classification efficiency and precision [60–62].

Class conditional independence refers to the notion that the effect of one attribute value on a particular class is unrelated to the value of any other attribute in the NBT process. Due to the application of the Bayes rule conditionally, NBT allows datasets to be trained in less time. This is because it treats all vectors as though they are independent [63]. equation (5) of the Bayes can be explained as follows:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (5)$$

Where $P(A|B)$ represent the conditional probability of A given B ; $P(B|A)$ showed the conditional probability of A given B . $P(A)$ explained the event probability of A . $P(B)$ explained the probability of event (B).

This classifier was selected because it (i) can be trained and classified rapidly, (ii) is resistant to noise in the form of inappropriate landscapes, (iii) is simple to implement and comprehend, and (iv) can be used effectively with very small quantities of training datasets [64].

The model begins with a series of probability estimates for each class, a calculation of the covariance and variance matrix, and the construction of the perception function [65]. Mapping and monitoring of forests in the KLR and NB were conducted using STATISTICA v. 7 [66–68].

Regression issues involving the prediction and mapping of continuing dependent variables are common applications of these tools' stochastic gradient boosting tree [69]. Then, after adjusting the settings, the system was optimized. This includes variables like the number of additive trees, the learning rate, the percentage of subsampling and etc. An ideal learning rate of 0.1 was used, with 185 additive trees and a maximum tree size of 5. Exactness of results may depend on these numbers [70]. The R package was used to carry out the classification process [71].

2.4. Ensemble of SVM, KLR and NBT

The stacking approach was used to construct the joint model, and a threshold of 0.9 was applied to identify forest pixels. Models are constructed for the SVM, NBT, and KLR, ML algorithms using the “glmnet” R package and the specified training datasets.

2.5. Model validation

Forest maps were normalized and compared pixel-by-pixel in order to assess the classifiers' efficacy, with values calculated for forest commission, total incorrect pixels, omission, precession, percentage of incorrect pixels, F1 and recall score. Once classification has been completed, it is crucial to assess classifier efficacy. Precision, accuracy, recall, and the F1 score were all shown to be useful measures of confusion, but the F1 score has gained the most traction in the research [24,72]. Precision, accuracy, F1 score and recall may all be broken down into its component parts by looking at these four metrics: True Negative (TN), True Positive (TP), False Negative (FN), and False Positive (FP). The following equations 6–10 can be used to determine the precision, accuracy, F1 score and recall:

$$\text{Accuracy} = \text{TP} + \left(\frac{\text{TN}}{\text{TP}}\right) \text{FP} + \text{FN} + \text{TN} \quad (6)$$

$$\text{Kappa} = p_o - \frac{p_e}{1} - p_e \quad (7)$$

where p_o is the observed agreement ratio and p_e is the expected agreement

$$\text{Precision} = \frac{\text{TP}}{\text{TP}} + \text{FP} \quad (8)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}} + \text{FN} \quad (9)$$

$$\text{F1} = 2 \times \text{precision} \times \frac{\text{recall}}{\text{precision} + \text{recall}} \quad (10)$$

where TP denotes a true positive result, FP a false positive, and FN is the false negative result. Free and open-source R 4.0.0 was used to conduct the performance analysis of the ML algorithms.

2.6. Training data collections

Improved classification accuracy can be achieved by using training datasets with higher spatial resolutions, but this can only be achieved by careful attention paid to the training data collecting process [73]. This technique creates training sample sizes that are proportional to the size of the classes by segmenting the population into similar subsets. Because of its proven effectiveness in lowering both bias and error rates, we choose to utilize this technique [53].

The classified images were validated using Google Earth (GE) and GPS points. The random sample point method within the Arc GIS spatial analyst was used to collect random points on the classified land uses of forests and non-forests images. These points were then translated and superimposed on GE into a kml format. Visual interpretation, expert experience and the values of GE points were evaluated. 1114 spatial training samples were obtained from a GPS field survey. The accuracy of classified images was measured using 70% (780) for forestmapping and training and 30% (334) for testing (validation) GPS points. Finally, comparing random samples and GPS with GE images was used to establish the uncertainty matrix. The uncertainty matrix was used to determine the specificity of the SVM land use/cover classification. We have used the Kappa Coefficient (K) as a predictor to confirm the qualitative consensus, either positive or negative, between graded samples and ground-truth points. The relative change of the classifier over ground-truth samples of land-use groups is usually determined from a mathematical evaluation.

2.7. Change detection

When comparing two classed images, the ID algorithm finds the difference in their DN values (1990–2000, 2000–2010, 2010–2017,

1990–2017, and 2013–2017). This can be used as a gauge of forest transformation across the KPK. The input parameters for change monitoring are the number of classes, the threshold value, and the type of change detection. Based on the type of remote sensing data, a threshold value is used to determine where in each class the forests have changed [74].

For simple difference, the cutoffs are uniformly distributed between (+1) and (−1). The first ($n/2$) classes are represented by the negative changes (positive value), and the latter ($n/2$) classes by the positive changes. Normalizing the images (raster maps) by subtracting the image minimum yields the no-change class $[(n/2) + 1]$, which represents the middle class (Fig. 2). Positive changes were shown by red pixels in the final state image of the change detection maps, whereas negative changes were indicated by blue pixels (initial state image). The Envi v.5.4 program included a change detection tool that was used to track the KPK forest as it evolved over time.

3. Results

3.1. Evaluation of machine learning performance and optimal parameterization

Fig. 3a, b and c compare the accuracy of SVM, KLR, and NBT in terms of recall and F1. The highest F1 scores (0.95) were obtained using SVM and KLR, while the lowest F1 score (0.85) was found with NBT. SVM and KLR algorithms were shown to have slightly different F1 scores. It indicates that the disparity in overall incorrect pixels between SVM, KLR, and NBT is responsible for the disparity in precision and recall scores. When compared to NBT, SVM and KLR shown superior capacity in differentiating between sparse and dense forests. All models, however, had a diminished capacity to identify forests due to their F1 score (0.8), which reflected a decreased capacity to distinguish between isolated forests. This reduced precision is the result of both reduced discrimination ability between forest and non-forest areas and reduced reflectivity inside forest areas. In comparison to NBT's predictions, those made by the SVM and KLR cover a broader range of forest coverage (Fig. 3d). A total of 103,559 km² was forecast by SVM, whereas 103,129 km² was forecast by KLR. Nonetheless, the NBT's forecast forest cover is less extensive than what can be seen in Landsat satellite images.

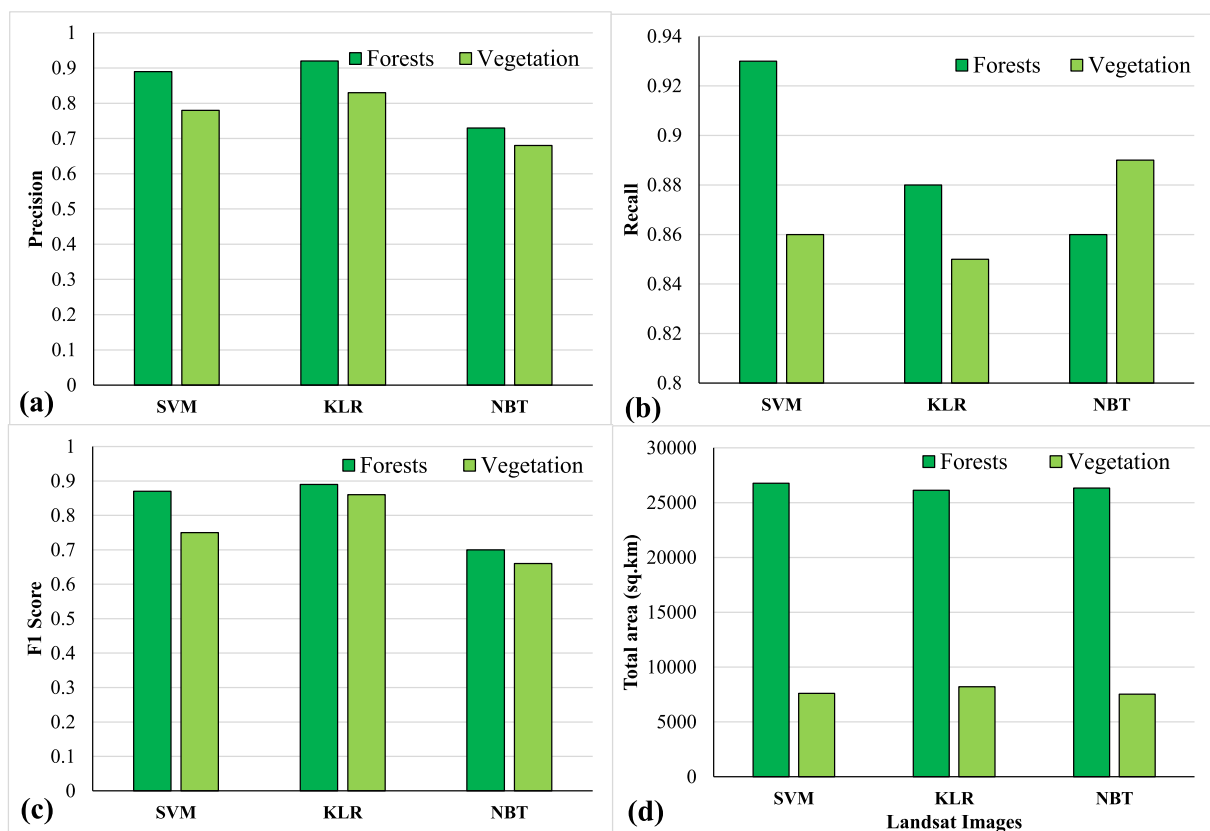


Fig. 3. Feature extraction and mapping of the total area (in Km²) from Landsat images of 2017 using SVM, KLR, and NBT models; (a) Precision, (b) Recall, (c) F1 Score, and (d) commulative area.

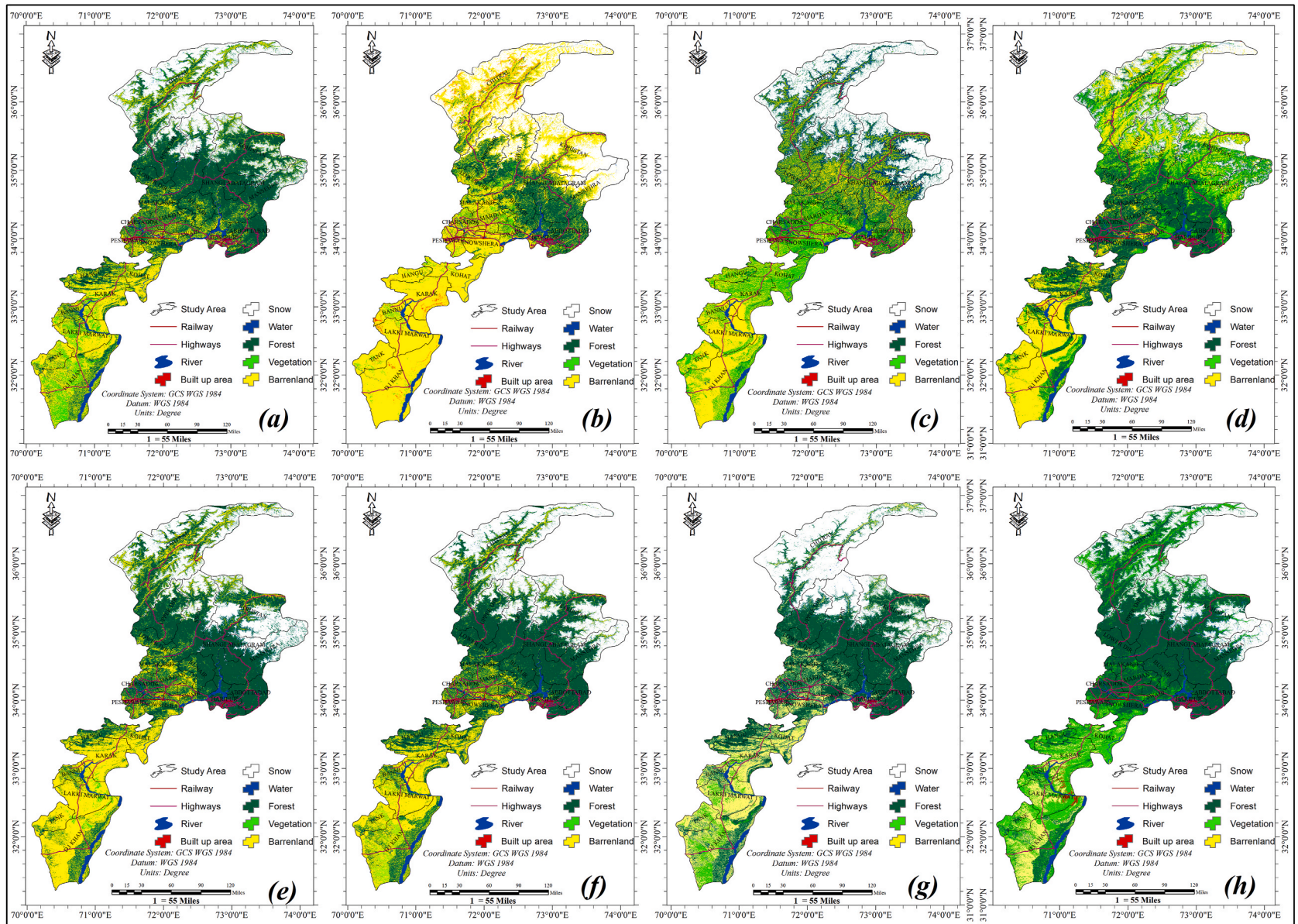


Fig. 4. Landuse/cover maps of (a) 1990, (b) 2000, (c) 2010, (d) 2013, (e) 2014, (f) 2015, (g) 2016, (h) 2017.

3.2. Landuse landcover classification

Fig. 4(a–h) illustrates the LULC classification obtained from SVM from 1990 to 2017 using Landsat 5, 7, and 8 satellite images. Mostly snow was displayed in northern areas of the KPK (Chitral) region, indicated in blue. Water bodies were displayed in the blue site, mainly in Abbottabad, Peshawar, Upper Dir, and Lower Dir. Indus River and Tarbella dam are primary freshwater sources in KPK. Forest area, barren land, built-up area, and vegetation were displayed in dark green, yellow, red, and light green colors. In all LULC maps from 1990 to 2017, snow areas were seen in most Chitral regions.

3.3. Change detection

During 1990 to 2000 forest area was decreased from 19,360 km² (26.0%) to 19,072 km² (26.6%), while barren land and built-up area increased from 19,801 km² (26.6) to 20,113 km² (27.0%) and 12,134 km² (16.4) to 12,525 km² (16.8%) respectively (Fig. 5a). From 2000 to 2010, forest area rapidly decreased due to urbanization and deforestation in KPK province. Forest area and vegetation was decreased 19,072 km² (25.6%) to 18,784 (25.2%) and 8601 km² (11.5%) to 8265 km² (11.1%), while barren land and built-up area was increased from 20,113 km² (27.0%) to 20,425 km² (27.4%) displayed in Fig. 5b. Fig. 5c clearly illustrates the forest decrease of almost 576 (almost 1%) of the total forest class of KPK from 2000 to 2010. Table 2 and Fig. 5d present variations in seasonal LULC distribution recorded at high accuracy using the SVM between 1990 and 2017. The accumulation of Forest areas in the central and

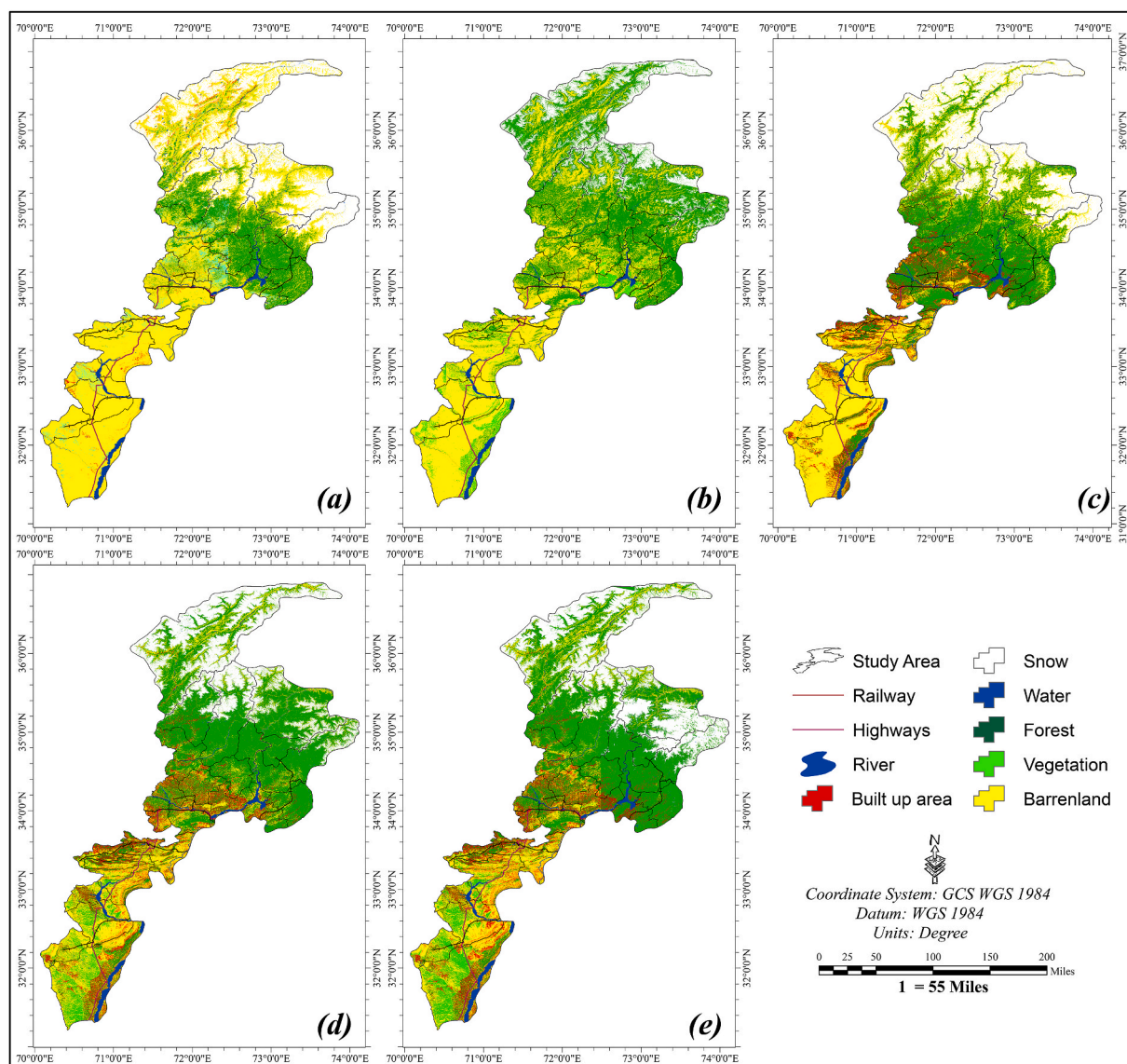


Fig. 5. Change detection map from (a) 1990–2000, (b) 2000–2010, (c) 2010–2017, (d) 1990–2017 and (e) 2013–2017.

northern regions is extended from 2013 to 2017. Before this time, the span of forest area was decreased from 1990 to 2010. Table 2 describes, built-up area, water, and vegetation area, Built-up area and forest area was increased from 12,213 km² (16.4%) to 13,173 km² (17.7%), 3041 km² (4.1%) to 3515 km² (4.7%) and 8937 km² (12%) to 9177 km² (12.3%) (Table 2). Meanwhile, snow, barren land and forest class decreased from 11,169 km² (15%) to 10,695 km² (14.4%), 19,801 km² (26.6%) to 19,321 km² (25.9%) and 19,360 km² (26.0%) to 18,640 km² (25%) respectively (Table 2). In the end, Fig. 5e displays the 2013 to 2017 forest class results. 2013–2017 showed that the area of forest cover increased from 18,640 km² (25.0) to 26,765 km² (35%) due to the one billion tree project. The change was 3.61 % as compared to the year 2015–2016. The overall change was 8.74 % forest area was increased from 2013-to 2017.

Forest cover was changed yearly to 25.39%, 27.23%, 30.52%, 34.13% for 2013, 2014, 2015, 2016, and 2017 respectively. Overall forest cover area from 2013 to 2017 was changed by 30–46% of the total area of the study region. The results of forest cover change as calculated by change detection have been shown in Fig. 5. The overall result of forest cover in 2013–2014 was 37,836 km² area covered with forest, and 111,206 km² area was shown non-forest of the total area of KPK. The change was 3.61 % as compared to the year 2015–2016. In 2014 and 2015, the forest-covered area was 40,578 km², and 108,464 km² area was non-forest covered. In 2015–2016, the forest-covered area was 45,483 km² and 103,559 km² was non-forest covered. Only 3 .29% of forest-covered were increased during 2015–2016.

4. Discussion

The study investigated the numerous land cover types in KPK Province in Pakistan. The classified results more administered the changes that had fallen between the identified classes.

4.1. Evaluation of the performance of the classifiers

Accurate forest mapping, classification, and 27 years change monitoring across ecosystems were made possible through the application and performance comparison of SVM, NBT and KLR, algorithms. SVM achieved the highest levels of accuracy, recall, and F1 score of any machine learning technique. Consequently, the SVM has a robust capacity to map forests across a variety of habitats. According to these findings, the SVM and KLR can identify dense forest sections that stand in for the entire forest. When compared to other methods, NBT's F1 score was the lowest (0.8), indicating its poorer forest detection performance (Fig. 4).

The F1 scores of SVM iterations were 0.93 and 0.90, respectively, whereas those of KLR iterations were 0.90 and 0.89. These results agreed well with other research that has found the SVM algorithm to be more accurate than the CART [75], RF [76], and maximum likelihood [77], who evaluated the efficacy of groundwater aid using machine learning methods to that of classical classifiers.

The results are consistent with those of [78]; who tested several classifiers for change monitoring and found that the SVM, which makes use of publicly available Landsat imagery, was the most effective. From multiple time-series of high-resolution remote-sensing images [79], implemented a change detection method using visual saliency and SVM. When it came to mapping forests, however, conventional classifiers actually had the opposite effect.

When compared to ML classifiers that used commercially accessible remote sensing data like LiDAR and WorldView-3, the accuracy of using publicly available Landsat images to recognize forests was on par [78,79]. SVM's total accuracy only reached 82%, despite the excellent spatial resolution of the LiDAR, WorldView, and Rapid Eye images [75]. In LULC classification, other investigations verified that the Rotation of Forest (RoF) approach, which is less sensitive than the SVM algorithm, was more accurate than the Canonical Correlation Forest (CCF) [34,75].

The suggested method significantly improves the efficiency of both linear and non-linear classifiers and constitutes a departure from traditional forests classes and LULC. With proper tuning, ML algorithms can improve classification quality and robustness [80, 81]. Processing massive volumes of remote sensing data quickly and cheaply was made possible by the widespread availability and simple implementation of ML, as well as the free availability of Landsat images [78,79].

4.2. Re-forestration of change detection

Our findings indicate that Landsat data combined with an SVM classification technique allows the LULC to be highly OA. Relevant satellite data is essential for achieving high accuracy levels. Despite a comparatively higher resolution, Landsat data is constrained not

Table 2
LULC from 1990 to 2017.

Class Name	Area in Km ² (Percentage)							
	1990	2000	2010	2013	2014	2015	2016	2017
Snow	11,169 (15.0)	11,122 (14.9)	10,652 (14.3)	10,695 (14.4)	9962 (13.4)	10,627 (14.3)	8850 (11.9)	8703 (11.7)
Water	3041 (4.1)	3088 (4.1)	3558 (4.8)	3515 (4.7)	3598 (4.8)	2116 (2.8)	3254 (4.4)	2932 (3.9)
Barren Land	19,801 (26.6)	20,113 (27.0)	20,425 (27.4)	19,321 (25.9)	18,491 (24.8)	16,137 (21.7)	15,204 (20.4)	14,377 (19.3)
Built up area	12,213 (16.4)	12,525 (16.8)	12,837 (17.2)	13,173 (17.7)	13,480 (18.1)	13,730 18.4	13,974 (18.8)	14,134 (19.0)
Forest area	19,360 (26.0)	19,072 (25.6)	18,784 (25.2)	18,640 (25.0)	19,196 (25.8)	21,382 (28.7)	24,101 (32.3)	26,765 (35.9)
Vegetation	8937 (12.0)	8601 (11.5)	8265 (11.1)	9177 (12.3)	9794 (13.1)	10,529 (14.1)	9138 (12.3)	7610 (10.2)

to promptly collect geospatial data, limiting its suitability for flood monitoring and mapping [82,83]. However, two Landsat satellite paths (150 and 151) enabled the acquisition of high-temporal resolution Landsat data (16 days) in the study area, compared to a single temporal resolution of 15 days for the site [84]. Cloud cover presence will restrict the availability of images of flood instances. Compared to optical satellites, SAR (Sentinel 1,2,3) and RADAR satellites can quickly penetrate clouds and acquire images (Landsat, MODIS) [82]. Variations that occurred during the period 2013–14 showed that forests had increased by 2.98%. From 2014 to 2015, the forest was the only prevailing land cover, with an area of 19,196 km² in 2014. However, it was enlarged to 21,382 km² in 2015 (Fig. 6).

The subsequent prevalent land cover in terms of terrestrial size was barren land that ensured a size reduction in 2015, but in 2014 it became the most dominant land cover with a total size of 18,491 km². The increase in the forest over the period confirmed the report by Bhaduri [85]. Change detection for 2014 to 2015 showed that forest area increased by 11.39%, making almost 2186 km². In this period, the forest-covered area was 27.23% of the total area of the study region. Almost 2719 km² of the new area was included in the forest class, almost 12.72% [86]. In this period, the forest-covered area was 30.52% of the total area of the study region. During the period 2016–17, the forest was the only most prevalent land cover, with an area of 24,101 km² in 2016. It had improved to 26,765 km² in 2017. But in 2016, it emerged as the most dominant land cover, with an overall area of 15,204 km² [87]. Forest area was increased to 11.05%, making almost 2664 km². In this period, the forest-covered area was 34.13% of the total area of the study region (Fig. 6).

The forest cover change map for 2013, 2014, 2015, 2016, and 2017 are shown in Fig. 6, respectively. For the year 2013, the map indicates that only a few districts of KPK were covered with forest, and a deficient proportion of forest cover was observed. During 2014, an increase in forest cover area was observed, although less. In 2015, the Government of Kpk made policies to improve the forest cover area and control deforestation. During 2016 and 2017, a significant increase in forest cover area was observed, likely due to the initiative of the tsunami of billion tree project in KPK province [42].

In this analysis, various products, sensors, and algorithms were compared to show the uncertainty of estimating forests. Based on the fact that there was an increase in the forest area, this paper suggested using different methods to secure forest areas. We developed a new way for reforestation and forest monitoring for the “One Billion Tree Project” plan in KPK, Pakistan. According this period, 2013–2017, almost 8125 km² area was included in the forest class [42].

The long-term Landsat-based forest maps generated in this research provide accurate and updated estimates of forest cover in KPK, Pakistan. Our findings, however, only illustrate the general dynamics of forest cover without details on forest composition, tree height, or the average age of observed forests. Optical sensors, including canopy closing and density, are more susceptible to horizontal tree structures. Vertical structure parameters are difficult to obtain or extract using only the visible spectral bands of Landsat images. On the other hand, appropriate field investigation samples are necessary to identify low and high-density forests or young and mature forests. However, in KPK, Pakistan, we have minimal field samples with these attributes. The combination of optical and microwave data and the GEE application now provides an excellent opportunity to obtain exact forest maps on a regional or global scale. For better estimates of forest areas and their transitions, image fusion and robust classification algorithms should be considered a priority.

Furthermore, owing to the “One Billion Tree” schemes in KPK, Pakistan, our analysis demonstrated a slight net loss of forests and some afforestation. However, proper identification of forest quality or the reaction of vegetation to changing climate and human activity is rarely recorded in this area. Future studies should focus on better and more accurate databases to have credible policy-making knowledge eventually to explore these issues.

5. Conclusions

This research proposed a unified strategy for integrating SVM learning with high-resolution spatial training data, SVM optimal parameterization, and post-classification improvement of the resulting forests and non-forests maps. Sample node value of 7, tree depth value of 0, and tree in forest value of 75 were found to be optimal settings. The results showed that the integration strategy, which makes use of Landsat imagery, is able to distinguish between forest and non-forest regions, with an increased overall accuracy of 6.7% compared to supervised classifications. The findings also revealed that KPK woods had expanded everywhere throughout the course of the last 27 years.

One of the approaches had been used successfully using multispectral and multi-temporal satellite imagery with a resolution of 30 m. It was observed that the imagery should be captured in middle infrared and near-infrared bands of the spectrum for high detection rates. Furthermore, the current study proved that vegetation and snow effectively determine forest cover change. Our findings, forest area, vegetation and snow area was decreased from 19,360 km² (26.0%) to 18,784 (25.2%), 8937 km² (12.0%) to 8,601 km² (11.5%), and 11,169 km² (15.0%) to 10,652 km² (14.3%), while barren land built-up area and water area was increased from 19,801 km² (26.6%) to 20,113 km² (27.0%), 12,213 km² (16.4%) to 12,525 km² (16.8%), and 3041 km² (4.1%) to 3558 km² (4.8%) in 1990–2010. Forest class rapidly increased due to one billion projects from 18,640 km² (25.0%) to 26,765 km² (35.9%), increasing almost 8125 km² from 2013 to 2017.

Further study will be conducted using multi-temporal Landsat data from 1990 to 2017 for identification of the spatio-temporal distribution of forest cover area according to forest type and using Cellular Automata Markov Chain analysis for future prediction of forest cover change. The study demonstrated concurrent reforestation in barren land areas. It has improved the methods to sustain the forest and RS and GIS in everyday forestry organization practices in the KPK area. This research would be highly effective, especially at the decision-making level.

Author contribution statement

Aqil Tariq: Conceived and designed the experiments; Performed the experiments; Analyzed and interpreted the data; Contributed

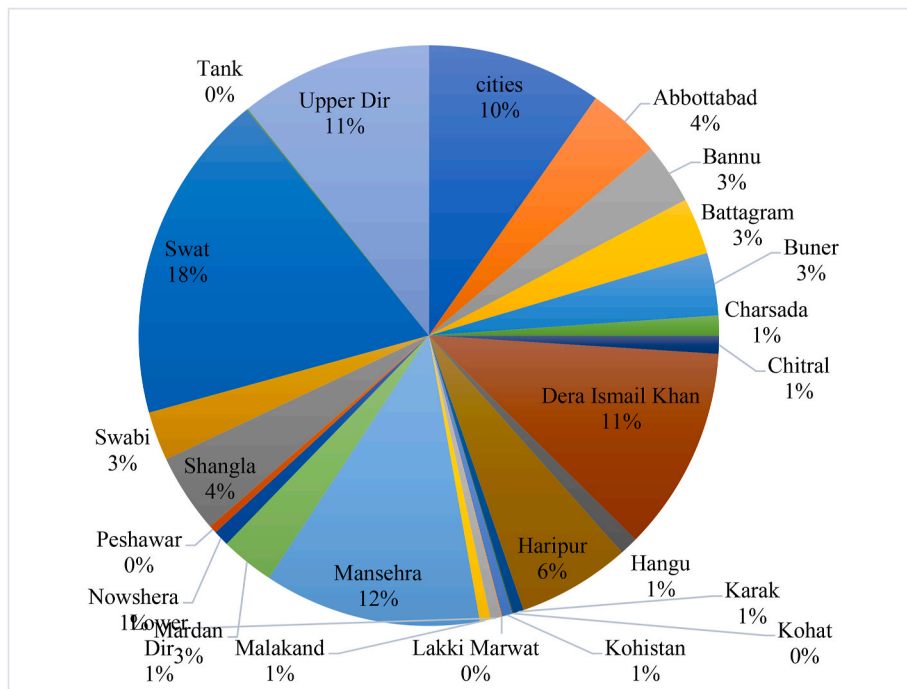


Fig. 6. Re-forestation map from 2013 to 2017.

reagents, materials, analysis tools or data; Wrote the paper. Yan Jiango, Qingting Li, Jianwei Gao, Walid Soufan, Khalid F. Almutairi, Muhammad Habib-ur-Rahman and Linlin Lu; Contributed reagents, materials, analysis tools or data; Wrote the paper.

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Data availability statement

Data included in article/supp. Material/referenced the in article.

Declaration of interest's statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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